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SUPERSEDING
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MILITARY SPECIFICATION

VIBRATORS, INTERRUPTER AND SELF-RECTIFYING

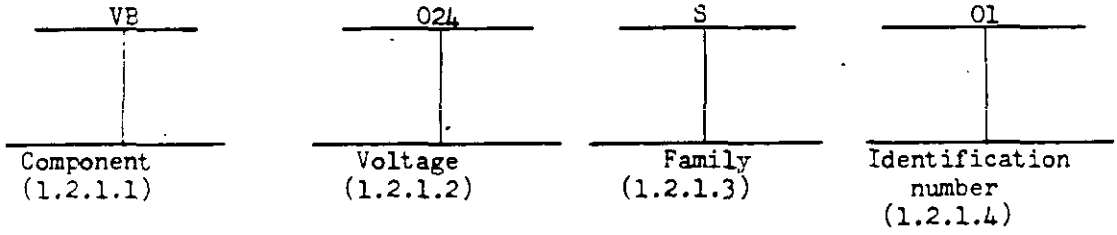
This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers interrupter and self-rectifying vibrators, designed to operate from a direct-current (dc) source, for use in electronic equipment.

1.2 Classification.

1.2.1 Type designation. The type designation of interrupter and self-rectifying vibrators shall be in the following form, and as specified (see 3.18 and 6.1):



1.2.1.1 Component. Vibrators are identified by the two-letter symbol "VB".

FSC 6130

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1.2.1.2 Voltage. The nominal dc input voltage is identified by a three-digit number. When the voltage is less than 10, it is preceded by two zeros; when the voltage is greater than nine but less than 100, it is preceded by one zero.

1.2.1.3 Family. The family is identified by a single letter denoting whether the vibrator is an interrupter (N) or is self-rectifying (S).

1.2.1.4 Identification number. The identification number indicates specific vibrators within a particular voltage-family series. The number, assigned serially, comprises two digits; if the number is less than 10, it is preceded by a zero.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-513	Plastic Sheet, Laminated, Thermosetting, Paper-Base, Phenolic-Resin
HH-I-538	Insulation, Electrical, Pasted Mica
PPP-B-566	Boxes, Folding, Paperboard
PPP-B-636	Box, Fiberboard
PPP-B-676	Boxes, Set-Up,
PPP-F-320	Fiberboard, Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes
PPP-P-291	Paperboard, Wrapping, Cushioning
PPP-T-45	Tape, Gummed, Paper, Reinforced and Plain, For Sealing and Securing
PPP-T-76	Tape, Pressure-Sensitive Adhesive Paper, Water Resistant (For Carton Sealing)
PPP-T-97	Tape, Pressure-Sensitive, Adhesive, Filament Reinforced

MILITARY

MIL-I-10	Insulating Materials, Electrical, Ceramic, Class L
MIL-M-14	Molding Plastics and Molded Plastic Parts, Thermosetting

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MIL-P-116	Preservation, Methods of
MIL-V-173	Varnish, Moisture and Fungus-Resistant (For the Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-W-583	Wire, Magnet, Electrical
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-C-45662	Calibration System Requirements

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U. S. Military Property
MIL-STD-147	Palletized Unit Loads (40 in. x 48 in. 4-Way Partial and 4-Way Pallets)
MIL-STD-202	Test Methods for Electronic and Electrical Component Parts
MIL-STD-454	Standard General Requirements For Electronic Equipment

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

3. REQUIREMENTS

3.1 General requirements. Vibrators furnished under this specification shall be a product which has passed the requirements of this specification. This specification can also be applied as a procurement document for vibrators other than those specified in the detail requirement paragraph (see 3.18) when used in conjunction with another document (e.g. service drawing, or other detail procurement document) giving detail requirements and listing additions and exceptions to the requirements and tests specified herein. In the event of any conflict between the requirements of this specification and the detail requirements, the latter shall govern. (See 6.1 and 6.1.1)

3.2 Preproduction samples. Preproduction samples are required, shall be fabricated by the contractor, and shall be subjected to the preproduction inspection specified in 4.4. The preproduction samples shall be assembled using parts, materials,

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and processes that will be employed in production, and shall be fabricated and assembled in a manner similar to that to be used in production.

3.2.1 Reference standards. After preproduction samples have been approved by the Government, they shall be kept intact in custody of the Government inspector until released by him. They shall be used as reference standards to resolve any differences of opinion regarding interpretation of requirements.

3.3 Material.

3.3.1 Plastic.

3.3.1.1 Laminated. Laminated plastic material shall be in accordance with Specification L-P-513.

3.3.1.2 Molded. Molded plastic material shall be in accordance with Specification MIL-M-14.

3.3.2 Ceramic. Ceramic material shall be in accordance with Specification MIL-I-10.

3.3.3 Mica. Mica material shall be in accordance with Specification HH-I-538.

3.3.4 Magnet wire. Magnet wire shall be in accordance with Specification MIL-W-583.

3.3.5 Tropicalization. Fungus inert and inherently noncorrosive materials shall be used. Materials which are nutrients for fungi shall be treated with a coating in accordance with MIL-V-173. Other materials which are not inherently non-corrosive shall be neutralized so that no corrosion will occur in the completed unit.

3.4 Design and construction.

3.4.1 Base and connections. The vibrator base shall be of the size and pin configuration specified. Base connections shall be as specified. (See 3.18)

3.4.2 Container. The vibrator container (can) shall be of the size and shape specified. (See 3.18)

3.5 Seal. There shall be no flow of air bubbles from the vibrator can during the test specified in 4.6.2.

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3.6 Dielectric strength. Vibrators shall withstand the specified potential without damage, arcing, or breakdown. (See 4.6.3)

3.7 Starting. Vibrators shall start immediately, shall operate normally, and the output voltage shall be as specified, when tested in accordance with 4.6.4. (See 3.18)

3.8 Load. The output voltage, input current, and frequency shall be as specified. (See 3.18 and 4.6.5)

3.9 Temperature cycling. The output voltage shall be as specified, when vibrators are tested in accordance with 4.6.6. (See 3.18)

3.10 Overvoltage. Vibrators shall start immediately, shall operate normally, and the output voltage shall be as specified, when tested in accordance with 4.6.7. (See 3.18)

3.11 Low-temperature starting and operating. Vibrators shall start immediately, shall operate normally, and the output voltage shall be as specified, when tested in accordance with 4.6.8. (See 3.18)

3.12 High-temperature starting and operating. Vibrators shall start immediately, shall operate normally, and the output voltage shall be as specified, when tested in accordance with 4.6.9. (See 3.18)

3.13 Moisture resistance (when applicable). The dielectric strength and starting shall be as specified in 3.6 and 3.7, respectively, when vibrators are tested in accordance with 4.6.10.

3.14 Vibration. There shall be no evidence of mechanical damage to the vibrator, and the output voltage shall be as specified, when vibrators are tested in accordance with 4.6.11. (See 3.18) Dents in the can which do not cause rupture of the can shall not be cause for rejection.

3.15 Shock (when applicable). There shall be no evidence of mechanical damage to the vibrator and the output voltage shall be as specified, when vibrators are tested in accordance with 4.6.12. Dents in the can which do not cause rupture of the can shall not be cause for rejection.

3.16 Life. Vibrators shall start immediately, shall operate normally, and the output voltage shall be as specified, when tested in accordance with 4.6.13. (See 3.18)

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3.17 Additional requirements.

3.17.1 Workmanship. Vibrators shall be manufactured and assembled in accordance with requirement 9 of Standard MIL-STD-454.

3.17.2 Soldering. Soldering shall be in accordance with requirement 5 of Standard MIL-STD-454.

3.17.3 Marking. Vibrator cans shall be marked in accordance with Standard MIL-STD-130, with "VIBRATOR," type designation, rated voltage, frequency, circuit schematic, manufacturer's name or symbol, and manufacturer's type number. No marking shall appear within 1/2 inch of the pin end of the can.

3.18 Detail requirements. Detail requirements applicable to individual vibrator types covered by this specification are specified in 3.18.1 for vibrator types VB006S01 and VB006S02, and in 3.18.2 for vibrator types VB024S01 and VB024S02.

3.18.1 Self-rectifying vibrator types VB006S01 and VB006SC2.

3.18.1.1 Physical requirements. Physical requirements shall be in accordance with Figure I.

3.18.1.2 Electrical requirements. Electrical rating, tests and requirements, and dielectric test shall be as specified in Tables I, II, and III, respectively, as follows:

Table I - Electrical rating (for information only)

Frequency	115 ± 7 Hz
Rated input voltage	6.3 vdc
Max. input current at rated input voltage	4.1 adc
Max. input voltage	8.0 vdc
Min. input voltage	5.0 vdc
Duty cycle	continuous
Life expectency	500 hrs.

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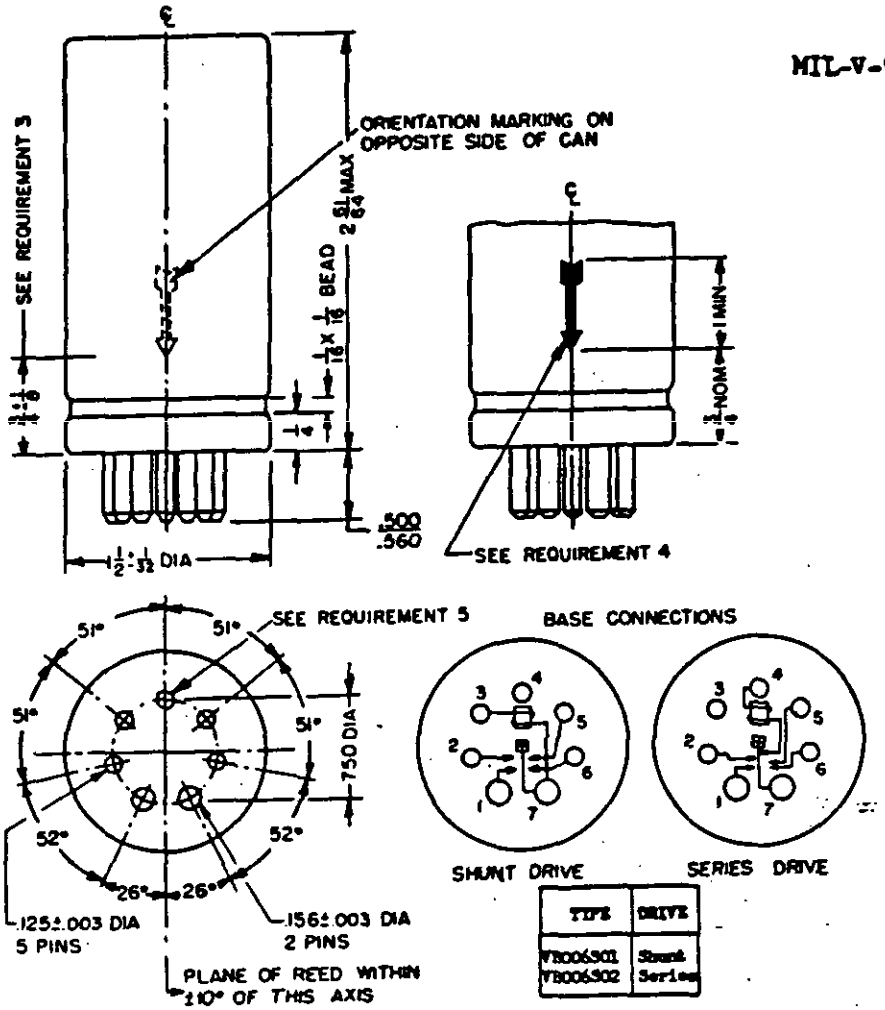


FIGURE I

1. All dimensions in inches.

2. Unless otherwise specified, tolerances are ± 1/64 on fractions, ± .005 on decimals, and ± 1/2° on angles.

3. To insure good electrical contact, do not treat lower portion of can when fungus-proofing is specified.

4. Rubber stamp or silk screen in a permanent contrasting color; shape optional, centered approximately between pins 1 and 7.

5. Pin 4 may be omitted on shunt-drive vibrators (type VB006S01). When pin 4 is wired in the equipment employing these vibrators, types VB006S01 and VB006S02 are directly interchangeable.

6. Contact pins shall be plated (silver or nickel) in accordance with Specification MIL-F-14072, type II.

7. Manufacturer's date-code number shall be legibly and permanently marked on the can.

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Table II - Test and requirements

Test	Input voltage	Output voltage
Starting	5.0	100 min.
Load (see Note 1)	6.3	136 ± 6
Temperature cycling	6.3	136 ± 6
Overvoltage	8.0	185 max.
Low temperature starting and operating	8.0	185 max.
	5.0	94 min.
High temperature starting and operating (see Note 2)	5.0	94 min.
	8.0	185 max.
Vibration	6.3	136 ± 6

NOTES:

1. Maximum input current: 4.1 adc
   Frequency: 115 ± 7 Hz.

2. The test temperature shall be 85° ± 3°C.
   for both preproduction testing and quality conformance testing.

Table III - Dielectric test

From Pin	To can connected to pins	Volts (RMS)
1	2, 5	500
2	1, 6, 7	500
5	1, 6, 7	500
6	2, 5	500
7	2, 5	500

3.18.1.3 Test circuits. All electrical tests shall be made using Power Supply PP-448/GRC dummy-loaded with a 1150-ohm, 22-watt resistor. Life tests shall be performed using Power Supply PP-448/GRC and Power Supply PP-109/GR as specified in 3.18.1.4(b). The dummy load for Power Supply PP-109/GR shall be that as shown in Figure II.

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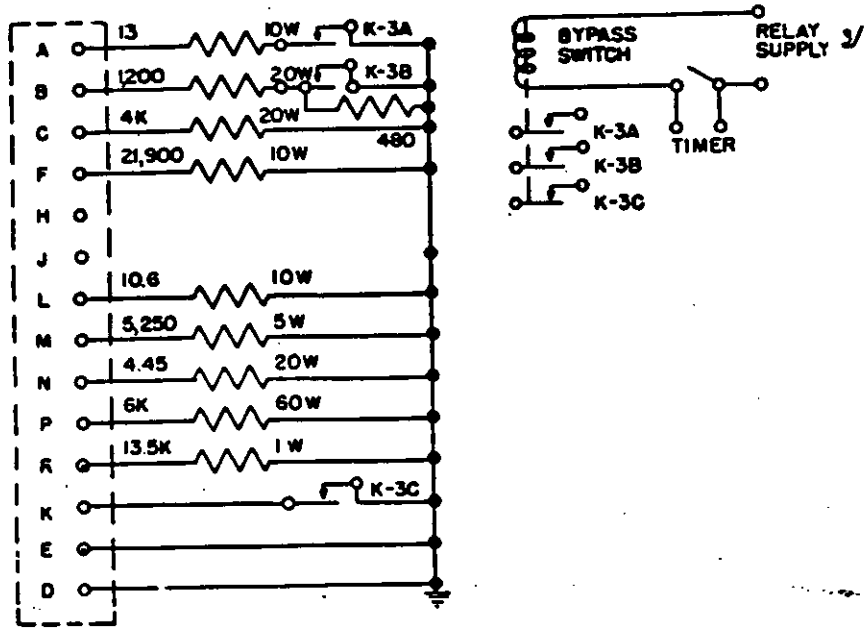


FIGURE II

1. All resistor values in ohms, ± 1%.

2. Switch S-1 of PP-109/GR shall be in the TRANS and RECEIVE Position and switch S-2 in the HIGH Position.

3. Relay K-3 is shown in the de-energized position and is used to change the load required for transmitting and receiving.

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3.18.1.4 Additional requirements. The following requirements shall also apply.

(a) Starting test. In addition to the starting test in Table II, the following shall apply. With 4.0 volts input applied, the vibrator shall attain full wave operation within 3 seconds. This test will become part of the starting test in test group I for preproduction tests, and in group A for acceptance tests. (See 4.6.4)

(b) Life tests. Life tests shall be made using Power Supplies PP-448/GRC and PP-109/GR. For preproduction tests, 6 vibrators shall be tested, 3 in each power supply. For group C acceptance tests, the sample size shall be 8, selected every 3 weeks; 2 vibrators shall be tested in PP-448/GR and 6 vibrators tested in PP-109/GR. The 8 vibrators selected shall be chosen at regular intervals during the preceding 3-week period. (See 4.6.13)

Life-test procedure when using Power Supply PP-448/GRC

(a) At 6.3 volts input, each vibrator shall be checked prior to life test. Output voltage shall be 136 ± 6 volts.

(b) At 7.5 volts input, ripple voltage shall be measured and shall not exceed 0.06 volt rms.

(c) At 7.5 volts input, test shall be run continuously for 500 hours.

(d) At the end of the first 24-hour period, initial output voltage measurement shall be made at 7.5 volts input.

(e) Output voltage shall be measured each working day and shall be within 10 percent of the initial measurement.

(f) At 7.5 volts input, after 500 hours life test, ripple voltage shall not exceed 0.10 volts rms.

Life-test procedure when using Power Supply PP-109/GR

(a) At 12.6 volts input, each vibrator shall be checked prior to life test. Output voltage at the following test points (see Figure II) shall be:

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Test Points	R	P	C	M	B
Volts	-27 ± 3	450 ± 25	150 ± 8	100 ± 7	85 ± 5

(b) At 15.0 volts input, test shall be run continuously for 500 hours, transmitting 2 minutes, receiving 6 minutes.

(c) At 10.0 volts input, vibrators shall be subjected to the starting test each working day for preproduction tests, and only once a week and at completion of life test for acceptance tests.

(d) At the end of the first 24-hour period, initial output voltage measurements shall be made at 15.0 volts input.

(e) Output voltage shall be measured each working day and shall be within 10 percent of the initial measurement.

NOTE: Failure of any of the vibrators to start shall constitute a life-test failure. The sample selected for group C acceptance tests will be considered satisfactory if, at the end of 500 hours of operation, the dc output of 8 vibrators has not decreased more than 10 percent.

(c) Seal test. Vibrators shall be immersed for 1 minute. (See 4.6.2)

(d) Moisture-resistance test. (See 4.6.10)

(e) Not for airborne use above 10,000 feet.

(f) Eighteen vibrators are required for preproduction tests unless otherwise specified in the contract or purchase order.

3.18.2 Self-rectifying vibrator types VB024S01 and VB024S02.

3.18.2.1 Physical requirements. Physical requirements shall be in accordance with Figure III.

3.18.2.2 Electrical requirements. Electrical rating, tests and requirements, and dielectric test shall be as specified in Tables IV, V, and VI, respectively, as follows:

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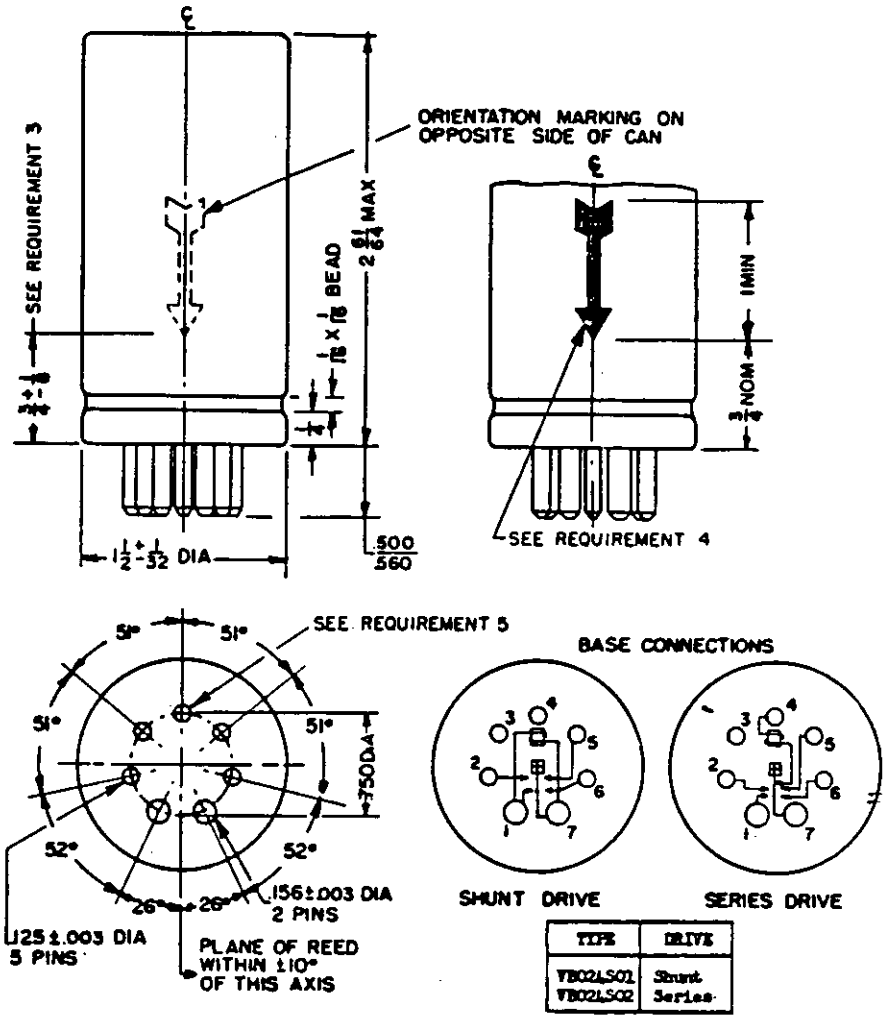


FIGURE III

1. All dimensions in inches.

2. Unless otherwise specified, tolerances are ± 1/64 on fractions, ± .005 on decimals, and ± 1/2° on angles.

3. To insure good electrical contact, do not treat lower portion of can when fungus-proofing is specified.

4. Rubber stamp or silk screen in a permanent contrasting color; shape optional, centered approximately between pins 1 and 7.

5. Pin 4 may be omitted on shunt-drive vibrators (type VB024S01). When pin 4 is wired in the equipment employing these vibrators, types VB024S01 and VB024S02 are directly interchangeable.

6. Contact pins shall be plated (nickel or silver) in accordance with Specification MIL-F-14072, type II.

7. Manufacturer's date-code number shall be legibly and permanently marked on the can.

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Table IV - Electrical rating (for information only)

Frequency	115 ± 7 Hz
Rated input voltage	25.2 vdc
Max. input current at rated input voltage	1.5 adc
Max. input voltage	32.0 vdc
Min. input voltage	20.0 vdc
Duty cycle	continuous
Life expectancy	500 hrs.

Table V - Tests and requirements

Test	Input voltage	Output voltage
Starting	20.0	100 min.
Load (see Note 1)	25.2	136 ± 6
Temperature cycling	25.2	136 ± 6
Overvoltage	32.0	185 max.
Low-temperature starting and operating	32.0	185 max.
	20.0	94 min.
High-temperature starting and operating (see Note 2)	20.0	94 min.
	32.0	185 max.
Vibration	25.2	136 ± 6

NOTES:

1. Maximum input current: 1.5 adc
   Frequency: 115 ± 7 Hz.

2. The test temperature shall be 85° ± 3°C for both preproduction testing and quality conformance testing.

Table VI - Dielectric test

From Pin	To can connected to pins	Volts (RMS)
1	2, 5	500
2	1, 6, 7	500
5	1, 6, 7	500
6	2, 5	500
7	2, 5	500

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3.18.2.3 Test circuits. All electrical tests shall be made using Power Supply PP-282/GRC dummy-loaded with a 1150-ohm, 22-watt resistor. Life test shall be performed using Power Supply PP-282/GRC and Power Supply PP-112/GR as specified in 3.18.2.4(b). The dummy load for Power Supply PP-112/GR shall be that as shown in Figure IV.

3.18.2.4 Additional requirements. The following requirements shall also apply.

(a) Starting test. In addition to the starting test in Table V, the following shall apply. With 16.0 volts input applied, the vibrator shall attain full wave operation within 3 seconds. This test will become part of the starting test in test group I for preproduction tests, and in group A for acceptance tests. (See 4.6.4)

(b) Life test. Life tests shall be made using Power Supplies PP-282/GRC and PP-112/GR. For preproduction tests, 6 vibrators shall be tested, 3 in each power supply. For group C acceptance tests, the sample size shall be 8, selected every 3 weeks; 2 vibrators shall be tested in PP-282/GRC and 6 vibrators test in PP-112/GR. The 8 vibrators selected shall be chosen at regular intervals during the preceding 3-week period. (See 4.6.13)

Life-test procedure when using Power Supply PP-282/GRC

(a) At 25.2 volts input, each vibrator shall be checked prior to life test. Output voltage shall be 136 ± 6 volts.

(b) At 30.0 volts input, ripple voltage shall be measured and shall not exceed 0.06 volts rms.

(c) At 30.0 volts input, test shall be run continuously for 500 hours.

(d) At the end of the first 24-hour period, initial output voltage measurement shall be made at 30.0 volts input.

(e) Output voltage shall be measured each working day and shall be within 10 percent of the initial measurement.

(f) At 30.0 volts input, after 500 hours life test, ripple voltage shall not exceed 0.10 volts rms.

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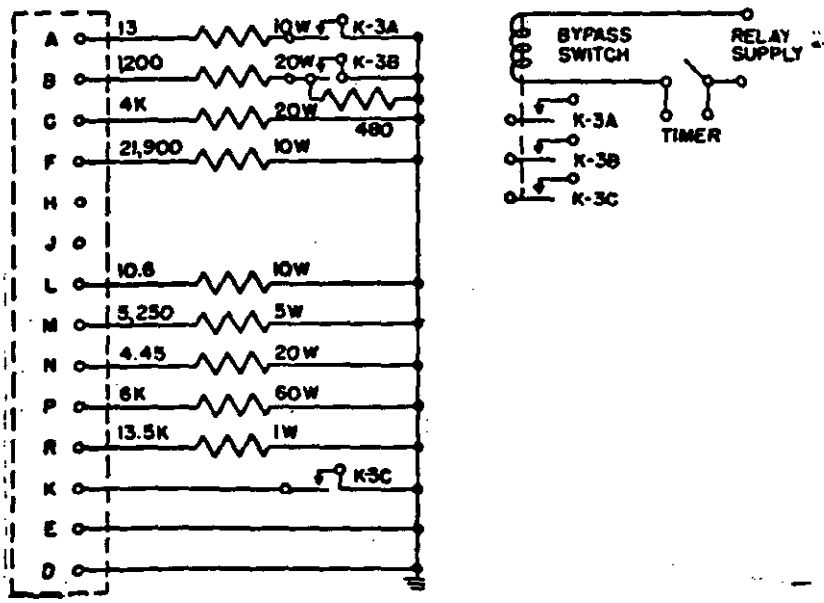


FIGURE IV

1. All resistor values in ohms, ± 1 percent.
2. Switch S-1 of PP-112/GR shall be in the TRANS and RECEIVE position and switch S-2 in the HIGH position.
3. Relay K-3 is shown in the de-energized position and is used to change the load required for transmitting and receiving.

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Life-test procedure when using Power Supply PP-112/GR

(a) At 25.2 volts input, each vibrator shall be checked prior to life test. Output voltage at the following test points (see Figure IV), shall be:

Test Points	R	P	C	M	B
Volts	-27 ± 3	450 ± 25	150 ± 8	100 ± 7	85 ± 5

(b) At 30.0 volts input, test shall be run continuously for 500 hours; transmitting 2 minutes, receiving 6 minutes.

(c) At 20.0 volts input, vibrators shall be subjected to the starting test each working day for preproduction tests, and only once a week and at completion of life test for acceptance tests.

(d) At the end of the first 24-hour period, initial voltage measurements shall be made at 30.0 volts input.

(e) Output voltage shall be measured each working day and shall be within 10 percent of the initial measurement.

NOTE: Failure of any of the vibrators to start shall constitute a life-test failure. The sample selected for group C acceptance tests will be considered satisfactory if, at the end of 500 hours of operation, the dc output of 8 vibrators has not decreased more than 10 percent.

(c) Seal test. Vibrators shall be immersed for 1 minute. (See 4.6.2)

(d) Moisture-resistance test. (See 4.6.10)

(e) Not for airborne use above 10,000 feet.

(f) Eighteen vibrators are required for preproduction tests unless otherwise specified in the contract or purchase order.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.3)

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4.1.1 Test equipment and inspection facilities. Test equipment and inspection facilities shall be of sufficient accuracy, quality and quantity to permit performance of the required inspection. Test equipment for Government verification shall be made available by the supplier. The supplier shall establish and maintain, to the satisfaction of the Government, a calibration system in accordance with Specification MIL-C-45662.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) Preproduction inspection (does not include preparation for delivery). (See 4.4)

(b) Quality conformance inspection.

(1) Quality conformance inspection of vibrators before preparation for delivery. (See 4.5)

(2) Quality conformance inspection of preparation for delivery. (See 4.7)

4.3 Inspection conditions. Unless otherwise specified herein, all measurements and tests shall be made at room ambient temperature, pressure, and humidity. When tests require that vibrators be placed in chambers, all related equipment shall be kept at ambient temperature and connected to the vibrators by use of the shortest possible lead lengths.

4.4 Preproduction inspection. Preproduction inspection shall consist of the examinations and tests specified in Table VII in the order shown. Unless otherwise specified, the number of samples of each type for which approval is requested shall be eighteen. After completion of the group I tests, the samples will be divided into two groups as shown in Table VII and subjected to the tests for their particular group.

Table VII - Preproduction inspection

Test	Rqt Para.	Test Para.
Test group I - all of sample		
Visual and dimensional inspection (external)	3.4, 3.17, 3.18	4.6.1
Seal	3.5	4.6.2
Dielectric strength	3.6	4.6.3
Starting	3.7	4.6.4
Load	3.8	4.6.5
Temperature cycling	3.9	4.6.6
Overvoltage	3.10	4.6.7

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Table VII - Preproduction inspection (Cont'd)

Test	Rqt Para.	Test Para.
Test group II - 2/3 of sample		
Low-temperature starting and operating	3.11	4.6.8
High-temperature starting and operating	3.12	4.6.9
Moisture resistance (when applicable)	3.13	4.6.10
Vibration	3.14	4.6.11
Shock (when applicable)	3.15	4.6.12
Test group III - 1/3 of sample		
Life	3.16	4.6.13

4.5 Quality conformance inspection. Quality conformance inspection of vibrators before preparation of delivery shall consist of group A, group B, and group C tests as specified in Tables VIII, IX, and X, respectively. Vibrators shall be presented for inspection in lots as defined by MIL-STD-105.

4.5.1 Group A inspection. This inspection, including sampling, shall conform to Table VIII and the inspection procedures of Standard MIL-STD-105, using the general inspection levels.

Table VIII - Group A inspection

Test	Rqt Para	Insp. Para.	AQL Major	AQL Minor
Visual and dimensional inspection	3.4, 3.17 and 3.18	4.6.1	1.0%	4.0%
Seal	3.5	4.6.2	1.0%	*
Dielectric strength	3.6	4.6.3)	1.0%	*
Starting	3.7	4.6.4)	for the	
Load	3.8	4.6.5)	group	

NOTE: *These inspections are Major category only.

4.5.2 Group B inspection. This inspection, including sampling, shall conform to Table IX and to the procedures for small-sample inspection of Standard MIL-STD-105, using the special inspection levels. Group B inspection shall normally be performed on inspection lots that have passed the group A inspection and on samples that have been subjected to and met the group A inspection requirements. Order of inspection shall be as shown in Table IX.

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Table IX - Group B inspection

Test	Rqt Para.	Test Para.	AQL
Temperature cycling	3.9	4.6.6)	6.5%
Low-temperature starting and operating	3.11	4.6.8)	for the
High-temperature starting and operating	3.12	4.6.9)	group
*Vibration	3.14	4.6.11	10.0%

* Sample units subjected to this inspection shall not be delivered on the contract or order.

4.5.2.1 Group B sampling plans. The group B sampling plans, for the AQL's shown in Table IX, shall be as follows:

AQL	Inspection level
6.5%	S-4
10.0%	S-3

4.5.2.2 Disposition of sample units. With the exception of the sample vibrators subjected to the vibration test, sample vibrators subjected to all other group B tests may be delivered on the contract or order provided they are resubjected to and pass the group A inspection requirements.

4.5.3 Unacceptable lots. If a lot should not conform to the group A or group B inspection requirements, the supplier may rework the lot or screen out the defectives and resubmit it for quality conformance inspection. Resubmitted lots shall be kept separate from new lots and identified accordingly. The resubmitted lot shall be inspected using a tightened sampling plan in accordance with Standard MIL-STD-105.

4.5.4 Group C inspection. This inspection shall consist of the tests specified in Table X and shall be performed at regular intervals, as specified (see 3.18), on sample vibrators selected from lots that have passed the group A inspection requirements. Sample vibrators shall be selected in accordance with 4.5.4.1.

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Table X - Group C inspection

Test	Rqt. Para.	Test Para.
Group I		
Moisture resistance	3.13	4.6.10
Shock	3.15	4.6.12
Group II		
Life	3.16	4.6.13

4.5.4.1 Group C sampling procedure.

4.5.4.1.1 Group I. Two sample vibrators for each test shall be selected at random for quantities up to and including 5000 units; and one additional sample, for each test, shall be selected for each additional increment of 2500 units or less. The first samples selected shall be at the start of the contract from the first quality conformance inspection production lot.

4.5.4.1.2 Group II. Sample units shall be selected without regard to quality, from the quantity produced during the previous production period. The production period shall be a period of time equal to the specified duration of the life test (see 3.18), expressed in the nearest whole number of days. For vibrators of the same type to be life-tested in only one type of test circuit, two sample units shall be selected for quantities up to and including 5,000 units, produced during the same production period; one additional sample unit shall be selected for each additional increment of 2,500 units or less, produced during the same production period. For vibrators of the same type to be life-tested in more than one type of test circuit, the sampling procedure shall be as specified. (See 3.18)

4.5.4.2 Noncompliance. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on (1) the process and materials, as applicable and (2) all units of product which were manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded to the responsible technical activity designated in the contract through the Quality Assurance Representative. After corrective action has been taken, additional sample units shall be subjected

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to group "C" inspection (all inspections, or the inspections which the sample failed, at the option of the Government); however, final acceptance and shipment will be withheld until the group C inspection results have shown that the corrective action was effective. (See 6.2)

4.5.4.3 Disposition of group C sample units. Sample vibrators which have been subjected to group C inspection shall not be delivered on the contract or order.

4.6 Test procedures.

4.6.1 Visual and dimensional inspection. Vibrators shall be inspected externally to verify that the design and construction, physical dimensions, marking, and workmanship are in accordance with the applicable requirements. (See 3.4, 3.17, and 3.18)

4.6.2 Seal. Vibrators shall be immersed in any suitable bath maintained at 80° to 85°C. for the period specified. (See 3.18) An alternate method of test may be used to determine satisfactory sealing, provided the method is proved to be equivalent to that specified herein. Any vibrator which shows evidence of leakage may be given remedial treatment provided evidence is submitted to show that such remedial treatment is adequate. (See 3.5)

4.6.3 Dielectric strength. The specified alternating current (ac) voltage of commerical-line frequency shall be applied as specified. (See 3.18) In no case shall the test voltage be applied across the driving coil. The test voltage shall be applied for not less than 1 minute during preproduction tests, and for not less than 5 seconds nor more than 1 minute during quality conformance inspection. (See 3.6)

4.6.4 Starting. With vibrators connected to the specified test circuit, the specified input voltage shall be applied through a snap switch 10 times, except for groups A and C inspection where it shall be applied 3 times. (See 3.18) Vibrators shall be checked for immediate starting. Output voltage shall be determined as soon as possible, but not later than 2 minutes after each start. The interval between stopping and starting shall be at least 5 seconds. The vibrator shall not be jarred during this test. (See 3.7)

4.6.5 Load. With vibrators connected to the specified test circuit, the specified input voltage (or voltages) shall be applied, and the output voltage, input current, and frequency shall be determined. (See 3.8 and 3.18)

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4.6.6 Temperature cycling. Vibrators shall be subjected to the temperature cycle shown in Table XI for preproduction inspection and for quality conformance inspection for a total of five cycles performed continuously. The vibrators shall be held at each temperature for sufficient time to allow all parts to reach thermal stability. Vibrators shall then be connected to the specified test circuit, the specified input voltage shall be applied, and the output voltage shall be determined. (See 3.9 and 3.18)

4.6.6.1 Test chambers. Separate test chambers shall be used for exposure of vibrators to extreme temperatures. The chambers shall be of sufficient thermal capacity to maintain a temperature of not less than 75°C. during the initial period of step 1 when units are being placed in the oven, and not more than -45°C. during the initial period of step 3 when units are being placed in the cold chamber. Otherwise, the temperature of the chambers shall stay within the tolerances shown in Table XI.

Table XI - Temperature cycle

Step	Temperature °C
1	85^{+3}_{-0}
2	25^{+10}_{-5}
3	-55^{+0}_{-3}
4	25^{+10}_{-5}

4.6.7 Overvoltage. Vibrators shall be connected to the specified test circuit and the specified overvoltage input shall be applied for 1 hour. (See 3.18) The output voltage shall be as specified. (See 3.10)

4.6.8 Low-temperature starting and operating. Vibrators shall be placed in a cold chamber maintained at $-55^{\circ}\ ^{+0^{\circ}}_{-3^{\circ}}$ C. until thermal stability is reached.

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While in the test chamber, vibrators shall be connected to the specified test circuit, and the specified maximum input voltage shall be applied. Vibrators shall then be checked for starting. The output voltage shall be determined within 2 minutes after application of the input voltage to the circuit. Vibrators shall then be de-energized and allowed to return to chamber temperature and the above test repeated, applying the specified minimum input voltage. (See 3.11 and 3.18)

4.6.9 High-temperature starting and operating. Vibrators shall be placed in a chamber at the specified ambient temperature (see 3.18) until thermal stability is reached. While in the test chamber, vibrators shall be connected to the specified test circuit, and the specified minimum input voltage shall be applied. Vibrators shall then be checked for starting. The output voltage shall be determined within 2 minutes after application of the input voltage to the circuit. Vibrators shall then be de-energized and allowed to return to chamber temperature and the above test repeated, applying the specified maximum input voltage. (See 3.12 and 3.18)

4.6.10 Moisture resistance (when applicable). Vibrators shall be tested in accordance with method 106 of Standard MIL-STD-202. The following details and exceptions shall apply:

(a) Mounting. Vibrators shall be mounted in sockets and clamped on a metal base in a manner similar to that employed under normal operating conditions.

(b) Initial measurements. Dielectric strength and starting tests (see 4.6.3 and 4.6.4) shall be made after the initial drying period.

(c) Polarization. During steps 1 to 6, inclusive, a dc potential of approximately 100 volts shall be applied across the insulation of 50 percent of the units. The potential shall be applied between the coil, frame, and reed as a postive terminal and all other pins as a negative terminal. The potential shall not be applied across any conductive circuit which might exist in the vibrator. No potential shall be applied to the other 50 percent of the units.

(d) Final measurements. Upon completion of step 6 of the final cycle, vibrators shall be removed from the humidity chamber and shall be conditioned for a period of 4 to 24 hours at room ambient temperature, pressure and humidity. Vibrators shall then be subjected to the dielectric-strength and starting tests specified in 4.6.3 and 4.6.4, respectively. (See 3.13 and 3.18)

4.6.11 Vibration. Vibrators shall be tested in accordance with method 201 of Standard MIL-STD-202. The following details and exceptions shall apply:

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(a) Mounting. Vibrators shall be mounted in sockets and clamped on the vibration platform in a manner similar to that employed under normal operating conditions.

(b) Initial measurements. Not applicable.

(c) Motion. One hour in each of three mutually perpendicular directions (total of 3 hours), one of which is perpendicular to the plane of the reed.

(d) Final measurements. Following the vibration test, vibrators shall be connected to the specified test circuit, the specified input voltage shall be applied, and the output voltage shall be determined. (See 3.14)

4.6.12 Shock (when applicable). Vibrators shall be tested in accordance with method 202 of Standard MIL-STD-202. The following details shall apply:

(a) Mounting. Vibrators shall be mounted in sockets and clamped on a metal base in a manner similar to that employed under normal operating conditions.

(b) The number and directions of blows, the desired acceleration, and the electrical load if applicable, shall be as specified.

(c) Measurements after shock. Vibrators shall be connected to the specified test circuit, the specified input voltage shall be applied, and the output voltage shall be determined. (See 3.15)

4.6.13 Life. The test circuit, duty cycle, and duration of test shall be as specified. During the life test, vibrators shall be checked for starting as specified. The output voltage shall be determined daily. (See 3.16 and 3.18)

4.7 Quality conformance inspection of preparation for delivery. Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of section 5.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C, as specified. (See 6.1(c))

5.1.1 Level A.

5.1.1.1 Cleaning. Cleaning shall be accomplished in accordance with process C-1 of MIL-P-116.

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5.1.1.2 Drying. Drying shall be accomplished in accordance with the applicable procedure of MIL-P-116.

5.1.1.3 Unit packaging. Unit packaging shall be in accordance with the methods prescribed in MIL-P-116 and as specified herein.

5.1.1.3.1 Technical literature. Each technical literature shall be packaged method 1C-1.

5.1.1.3.2 Vibrators. Vibrators shall be packaged method III, as follows; cushion each vibrator by wrapping in paperboard conforming to PPP-P-291, type III. Secure the cushioning with gummed paper tape conforming to PPP-T-45, type III, grade A. Place each cushioned vibrator within close-fitting box conforming to PPP-B-566 or PPP-B-676, variety 2. Close the box in accordance with the appendix of the applicable box specification.

5.1.1.3.3 Intermediate package. Place a quantity of vibrators, bearing the same stock number, and packaged as specified in 5.1.1.3.2, within a close-fitting fiberboard box conforming to PPP-B-636, grade W6c. Gross weight of the intermediate package shall not exceed 20 pounds. Close the box as specified in the appendix of the box specification, in addition seal all seams and joints with pressure-sensitive tape, conforming to PPP-T-76.

5.1.2 Level C. Vibrators shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packing. Packing shall be level A or C as specified. (See 6.1(c))

5.2.1 Level A.

5.2.1.1 A quantity of vibrators packaged as specified in 5.1.1.3.3, shall be packed within a close-fitting box conforming to PPP-B-636, type CF, class weather-resistant, special requirements use. Box closure shall be as specified in the appendix of the specification. To facilitate palletization, fiberboard boxes shall be uniform in size and contain equal quantities of the packaged items to the greatest extent practicable.

5.2.1.2 Palletized load, vibrators. A quantity of containers packed as specified in 5.2.1.1 shall be placed on a pallet, load type 1, conforming to MIL-STD-147. A fiberboard cap shall be employed over the load having two sides extending down the stacked load at least 12 inches to accommodate marking requirements. The cap shall be fabricated of fiberboard conforming to PPP-F-320, class weather-resistant, W5s. The load shall be "bonded" to the pallet by strapping.

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5.2.1.3 Less than palletized load. When quantities per destination are less than a pallet load, containers specified in 5.2.1.1 shall be reinforced by pressure sensitive filament tape conforming to PPP-T-97, type IV as specified in the appendix to the box specification. No further packing shall be required.

5.2.2 Level C. Vibrators packaged as specified in 5.1, shall be packed as specified in 5.2.1, except that pallets, fiberboard caps shall be of the type, size and kind commonly used for the purpose and shall comply with the rules and regulation of the common carrier as applicable to the mode of transportation. Packing shall be accomplished in a manner that will afford adequate protection to the package and its contents during shipment from the supply source to the first receiving activity.

5.3 Marking. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification.

(b) Complete type designation. (See 1.2.1 and 3.18)

(c) Level of packaging and level of packing required for shipment (Level A, level B, or level C).

(d) The specific paragraphs of section 5 which are applicable to the particular procurement.

(e) Preproduction sample requirements.

6.1.1 Vibrators to which type designations have not been assigned. Vibrators to which type designations have not been assigned may be procured under this specification, with the following information substituted for item (b) of 6.1:

(a) Family. (See 1.2.1.3)

(b) Maximum input current at rated input voltage.

(c) Nominal dc input voltage. (See 1.2.1.2)

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(d) Reed frequency.

(e) Size and pin configuration of base, and base connections (diagram).

(f) Can shape and dimensions.

(g) Nomenclature and circuit diagram of equipment in which vibrator is to be used.

(h) Nominal output voltage.

(i) All applicable tolerances.

(j) Additional marking information. (See 3.17)

(k) Life expectancy.

(l) Plane of reed.

(m) Exceptions to this specification. The above list contains the minimum information needed for procurement of the vibrator.

6.2 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.4.2)

6.3 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

NOTE: Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

Custodians
Army - EL
Navy - EC
Air Force - 82

Review activities
Army - EL,
Navy - EC, SH
Air Force - 82, 85

User activities
Army - EL
Navy - EC, AS, OS, CG, MC

Preparing activity

ARMY - EL

Project Nr. 6130-0047

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

YES NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

NAME *(Printed or typed name and activity)* | DATE

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